

Enterprise Data Warehouse Design

1. Introduction

This document presents the design of a production-grade Enterprise Data Warehouse for a logistics ERP system. The objective is to build a scalable analytical system using dimensional modeling techniques such as Star Schema and Snowflake Schema. The data warehouse supports business intelligence reporting and advanced analytics.

2. ER Diagram (Conceptual Design)

The Entity Relationship (ER) Diagram represents the structure of the data warehouse, including fact tables and dimension tables. The design follows a star schema approach where fact tables are centrally connected to multiple dimension tables.

- Core Fact Tables:
 - Fact_Sales_Orders
 - Fact_Purchase_Orders
 - Fact_Shipments
 - Fact_Inventory_Movements
 - Fact>Returns
 - Fact_Goods_Receipt
 - Fact_Payments
 - Fact_Warehouse_Operations
- Dimension Tables:
 - Dim_Customer
 - Dim_Product
 - Dim_Supplier
 - Dim_Employee
 - Dim_Warehouse
 - Dim_Courier
 - Dim_Geography
 - Dim_Calendar
 - Dim_Currency
 - Dim_Sales_Territory

Relationships: Fact tables maintain many-to-one relationships with dimension tables. Each fact record links to descriptive attributes through foreign keys.

3. Data Warehouse Architecture

The data warehouse is designed using the Kimball dimensional modeling approach. It includes fact tables for transactional data and dimension tables for descriptive attributes. The architecture supports efficient querying and reporting.

4. Fact Tables Description

Fact tables store measurable business events.

- Fact_Sales_Orders: Stores sales transaction details.
- Fact_Purchase_Orders: Stores supplier purchase data.
- Fact_Shipments: Tracks shipment and delivery information.
- Fact_Inventory_Movements: Records stock movement.
- Fact>Returns: Tracks returned products.
- Fact_Goods_Receipt: Records received goods.
- Fact_Payments: Stores payment transactions.
- Fact_Warehouse_Operations: Tracks warehouse activities.

5. Dimension Tables Description

Dimension tables provide descriptive context.

- Dim_Customer: Customer details.
- Dim_Product: Product information.
- Dim_Supplier: Supplier details.
- Dim_Employee: Employee data.
- Dim_Warehouse: Warehouse details.
- Dim_Courier: Courier information.
- Dim_Geography: Location hierarchy.
- Dim_Calendar: Date and time attributes.
- Dim_Currency: Currency types.
- Dim_Sales_Territory: Sales regions.

6. Key Design Concepts

- Star Schema:
 - Central fact table connected to denormalized dimension tables.
- Snowflake Schema:
 - Normalized dimension tables to reduce redundancy.
- Surrogate Keys:
 - Artificial primary keys used for better performance.
- Slowly Changing Dimension (Type 2):
 - Maintains historical data using StartDate, EndDate, and Active flag.
- Degenerate Dimensions:
 - Attributes stored in fact tables such as OrderID.
- Bridge Tables:
 - Handles many-to-many relationships.
- Snapshot Fact Tables:
 - Captures data at specific time intervals.
- Aggregate Fact Tables:
 - Stores precomputed summaries for faster queries.

7. Implementation Tools

The data warehouse is implemented using Power BI:

- Power Query for ETL
- Data Model for relationships

8. Conclusion

The designed data warehouse provides a scalable and efficient system for analytics. Dimensional modeling ensures high performance and easy reporting.